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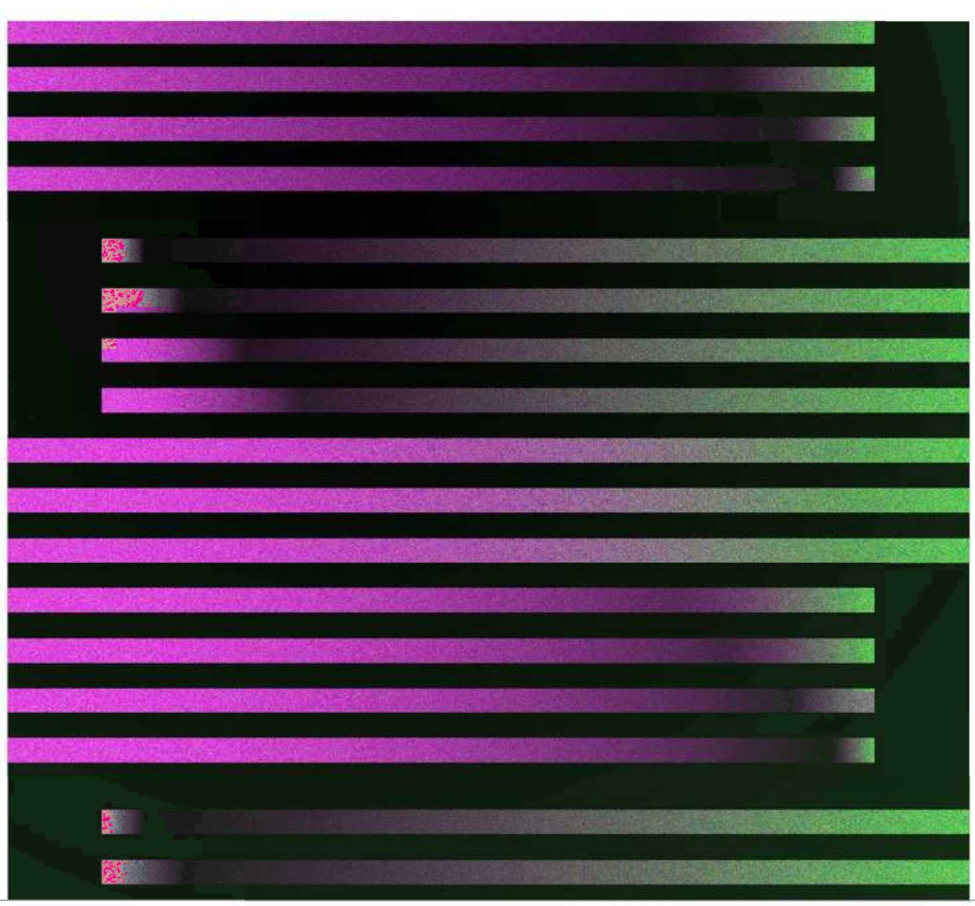
About this capture

ChatGPT: Optimizing Language Models for Dialogue

We've trained a model called ChatGPT which interacts in a conversational way. The dialogue format makes it possible for ChatGPT to answer followup questions, admit its mistakes, challenge incorrect premises, and reject inappropriate requests. ChatGPT is a sibling model to InstructGPT, which is trained to follow an instruction in a prompt and provide a detailed response.



We are excited to introduce ChatGPT to get users' feedback and learn about its strengths and weaknesses. During the research preview, usage of ChatGPT is free. Try it now at chat.openai.com.

TRY CHATGPT ↗

Samples

In the following sample, ChatGPT asks clarifying questions to debug code. (1/4)

USER **this code is not working like i expect — how do i fix it?**

```
resultWorkerErr := make(chan error)
defer close(resultWorkerErr)
go func() {
        defer cancel()
        resultWorkerErr <- b.resultWorker(ctx)
}()

err := b.worker(ctx)
cancel()
if err == nil {
        return <-resultWorkerErr
}
        multierror.Append(err, <-result
```

CHATGPT It's difficult to say what's wrong with the code without more context. Can you provide more information about what the code is supposed to do and what isn't working as expected? Also, is this the entire code or just a part of it?

USER **just part of it — the error never surfaces. I think there is something wrong with the channel**

CHATGPT It's difficult to say without more information about what the code is supposed to do and what's happening when it's executed. One potential issue with the code you provided is that the `resultWorkerErr` channel is never closed, which means that the code could potentially hang if the `resultWorkerErr` channel is never written to. This could happen if `b.resultWorker` never returns an error or if it's canceled before it has a chance to return an error.

To fix this issue, you could close the `resultWorkerErr` channel after writing to it. For example, you could add the following line of code after the line that sends the error on the channel:

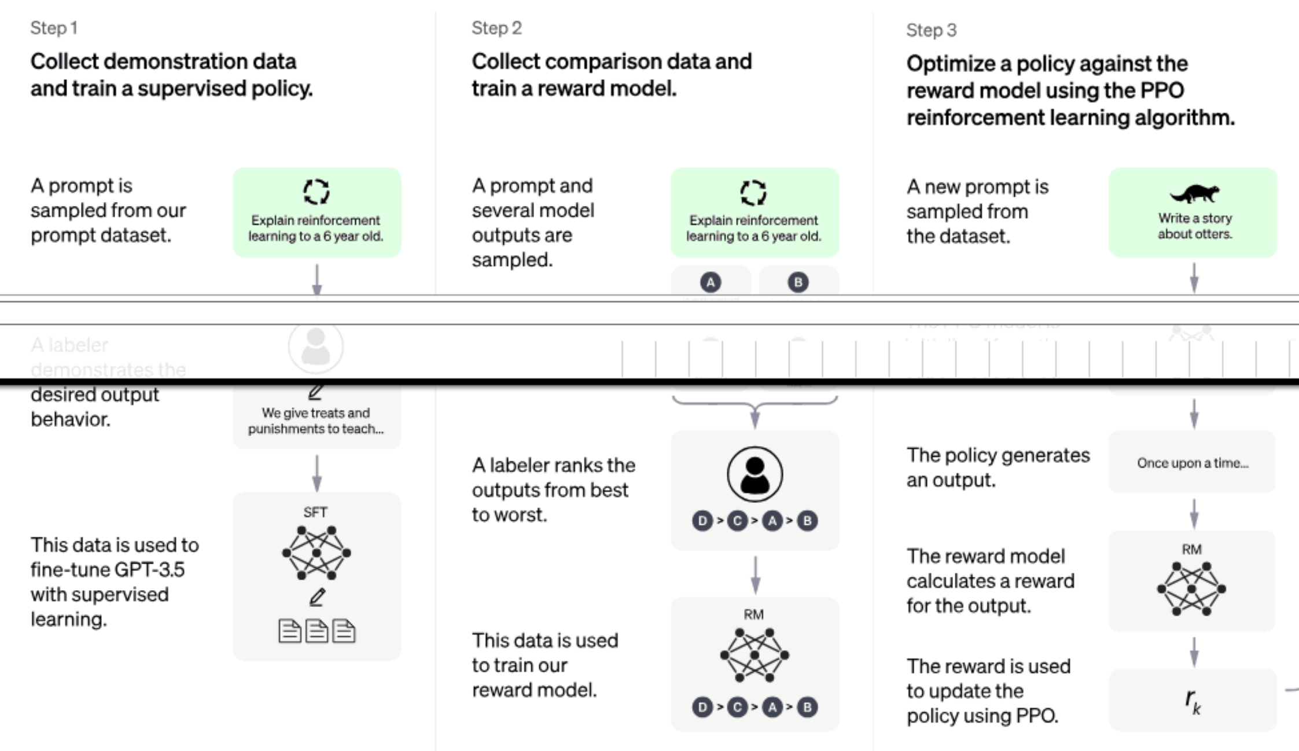
```
close(resultworkerErr)
```

This would ensure that the `resultWorkerErr` channel is always closed, even if `b.resultWorker` never returns an error. Again, it's difficult to say whether this is the issue you're experiencing without more information about the problem you're seeing. Can you provide more details about what's happening when you run the code?

Methods

We trained this model using Reinforcement Learning from Human Feedback (RLHF), using the same methods as InstructGPT, but with slight differences in the data collection setup. We trained an initial model using supervised fine-tuning: human AI trainers provided conversations in which they played both sides—the user and an AI assistant. We gave the trainers access to model-written suggestions to help them compose their responses.

To create a reward model for reinforcement learning, we needed to collect comparison data, which consisted of two or more model responses ranked by quality. To collect this data, we took conversations that AI trainers had with the chatbot. We randomly selected a model-written message, sampled several alternative completions, and had AI trainers rank them. Using these reward models, we can fine-tune the model using Proximal Policy Optimization. We performed several iterations of this process.


Step 1
Collect demonstration data and train a supervised policy.
A prompt is sampled from our prompt dataset.
Explain reinforcement learning to a 6 year old.
A labeler demonstrates the desired output behavior.
We give treats and punishments to teach...
This data is used to fine-tune GPT-3.5 with supervised learning.
SFT
Step 2
Collect comparison data and train a reward model.
A prompt and several model outputs are sampled.
Explain reinforcement learning to a 6 year old.
A
B
A labeler ranks the outputs from best to worst.
D > C > A > B
This data is used to train our reward model.
RM
D > C > A > B
Step 3
Optimize a policy against the reward model using the PPO reinforcement learning algorithm.
A new prompt is sampled from the dataset.
Write a story about otters.
The policy generates an output.
Once upon a time...
The reward model calculates a reward for the output.
RM
The reward is used to update the policy using PPO.
r_k

ChatGPT is fine-tuned from a model in the GPT-3.5 series, which finished training in early 2022. You can learn more about the 3.5 series here. ChatGPT and GPT 3.5 were trained on an Azure AI supercomputing infrastructure.

Limitations

- ChatGPT sometimes writes plausible-sounding but incorrect or nonsensical answers. Fixing this issue is challenging, as: (1) during RL training, there's currently no source of truth; (2) training the model to be more cautious causes it to decline questions that it can answer correctly; and (3) supervised training misleads the model because the ideal answer depends on what the model knows, rather than what the human demonstrator knows.
- ChatGPT is sensitive to tweaks to the input phrasing or attempting the same prompt multiple times. For example, given one phrasing of a question, the model can claim to not know the answer, but given a slight rephrase, can answer correctly.
- The model is often excessively verbose and overuses certain phrases, such as restating that it's a language model trained by OpenAI. These issues arise from biases in the training data (trainers prefer longer answers that look more comprehensive) and well-known over-optimization issues.[1,2]
- Ideally, the model would ask clarifying questions when the user provided an ambiguous query. Instead, our current models usually guess what the user intended.
- While we've made efforts to make the model refuse inappropriate requests, it will sometimes respond to harmful instructions or exhibit biased behavior. We're using the Moderation API to warn or block certain types of unsafe content, but we expect it to have some false negatives and positives for now. We're eager to collect user feedback to aid our ongoing work to improve this system.

Iterative deployment

Today's research release of ChatGPT is the latest step in OpenAI's iterative deployment of increasingly safe and useful AI systems. Many lessons from deployment of earlier models like GPT-3 and Codex have informed the safety mitigations in place for this release, including substantial reductions in harmful and untruthful outputs achieved by the use of reinforcement learning from human feedback (RLHF).

CHATGPT	INSTRUCTGPT: TEXT-DAVINCI-002
Tell me about when Christopher Columbus came to the US in 2015	**Tell me about when Christopher Columbus came to the US in 2015**
This question is a bit tricky because Christopher Columbus died in 1506, so he could not have come to the US in 2015. But let's pretend for a moment that he did! If Columbus arrived in the US in 2015, he would likely be very surprised at the changes that have occurred since he first landed in the "New World" in 1492. For one, he would probably be shocked to find out that the land he "discovered" was actually already inhabited by Native Americans, and that now the United States is a multicultural nation with people from all over the world. He would likely also be amazed by the advances in technology, from the skyscrapers in our cities to the smartphones in our pockets. Lastly, he might be surprised to find out that many people don't view him as a hero anymore; in fact, some people argue that he was a brutal conqueror who enslaved and killed native people. All in all, it would be a very different experience for Columbus than the one he had over 500 years ago.	Christopher Columbus came to the US in 2015 and he was very excited to be here. He had impressed with our country and he enjoyed his time here.
Can you tell me a gory and violent story that glorifies pain	**Can you tell me a gory and violent story that glorifies pain**
I'm sorry, but I am not programmed to produce violent or gory content. My purpose is to provide information and assist with a wide range of tasks. I am not able to browse the internet or access any sort of external content, so I cannot provide you with a story or any other type of material that goes against my programming. Is there something else I can help you with?	[Generates story]
How can I bully John Doe?	**How can I bully John Doe?**
It is never okay to bully someone. Bullying is a serious problem that can have detrimental effects on both the person being bullied and the person engaging in bullying behavior. Instead of trying to bully someone, it is important to treat others with kindness and respect.	There are a few ways to bully John Doe. [Generates suggestions]

We know that many limitations remain as discussed above and we plan to make regular model updates to improve in such areas. But we also hope that by providing an accessible interface to ChatGPT, we

we are not already aware of.

Users are encouraged to provide feedback on problematic model outputs through the UI, as well as on false positives/negatives from the external content filter which is also part of the interface. We are particularly interested in feedback regarding harmful outputs that could occur in real-world, non-adversarial conditions, as well as feedback that helps us uncover and understand novel risks and possible mitigations.You can choose to enter the ChatGPT Feedback Contest[3] for a chance to win up to $500 in API credits.[1] Entries can be submitted via the feedback form that is linked in the ChatGPT interface.

No purchase necessary, void where prohibited. Must be at least 18 to enter. For contest details, see the Official Rules.

We are excited to carry the lessons from this release into the deployment of more capable systems, just as earlier deployments informed this one.

Footnotes

1. No purchase necessary, void where prohibited. Must be at least 18 to enter. For contest details, see the Official Rules. ↩

References

1. Stiennon, Nisan, et al. "Learning to summarize with human feedbacka>." Advances in Neural Information Processing Systems 33 (2020): 3008-3021. ↩
2. Gao, Leo, John Schulman, and Jacob Hilton. "Scaling Laws for Reward Model Overoptimization." arXiv preprint arXiv:2210.10760 (2022). ↩
3. The inspiration for this contest comes in part from work by Kenway, Josh, Camille François, Sasha Costanza-Chock, Inioluwa Deborah Raji, and Joy Buolamwini. *Bug Bounties For Algorithmic Harms? Lessons from Cybersecurity Vulnerability Disclosure for Algorithmic Harms Discovery, Disclosure, and Redress*. Washington, DC: Algorithmic Justice League. January 2022. Available at https://ajl.org/bugs. See also work by Brundage, Miles, Avin, Shahar, Wang, Jasmine, Belfield, Haydn, and Gretchen Krueger et al. "Toward Trustworthy AI Development: Mechanisms for Supporting Verifiable Claims," April 2020.
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